UNIVERSITI TENAGA NASIONAL

The Energy University

CGEB4112 · ENERGY & COMPUTING · SECTION 1

Blockchain in the Oil & Gas Industry

GROUP 10

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How is blockchain changing that?

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Blockchain in the Oil & Gas Industry

01 How the oil business works

02 What blockchain is

03 Where it’s used

04 Real example: VAKT

05 Challenges & benefits

Three stages of oil

Petronas: all three



Upstream

Find · Drill · Produce



Midstream

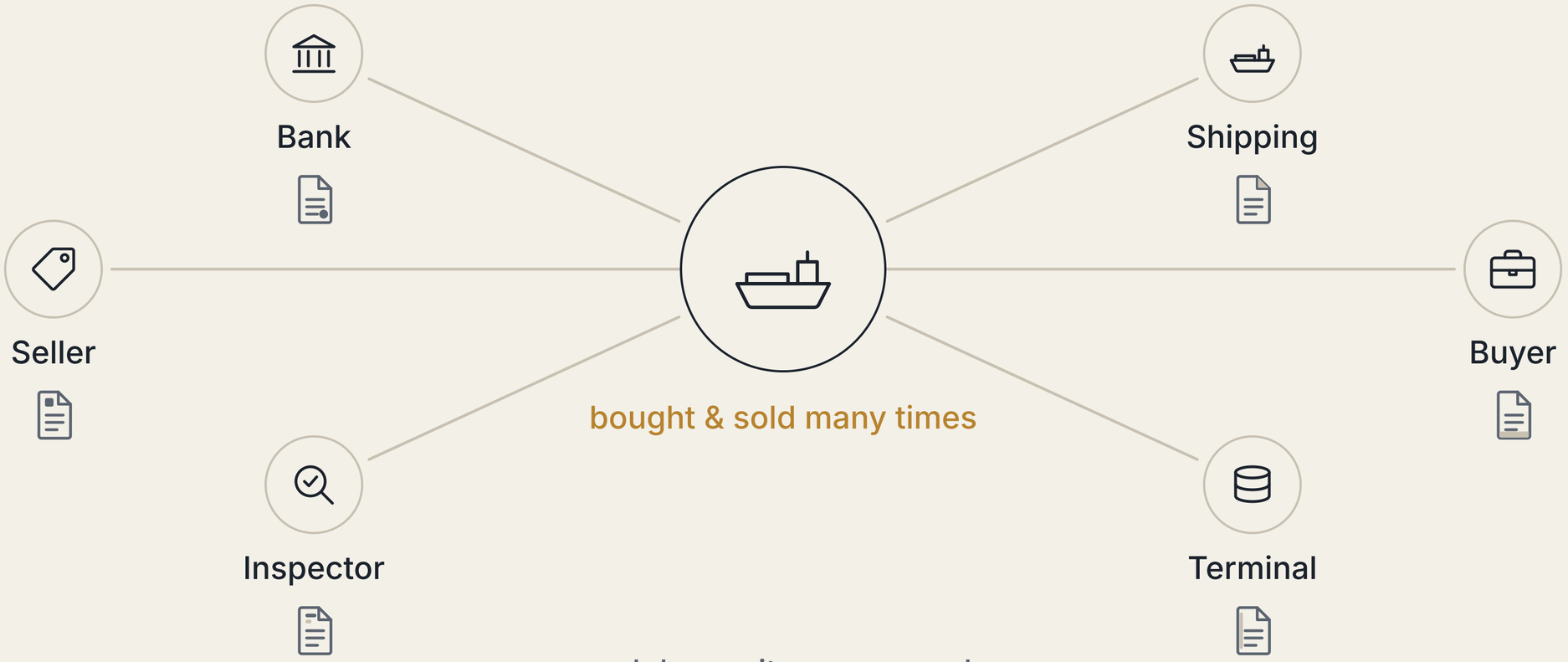
Transport · Store



Downstream

Refine · Sell

One cargo, many hands



each keeps its own records

No single version of the truth.

Blockchain = one shared record

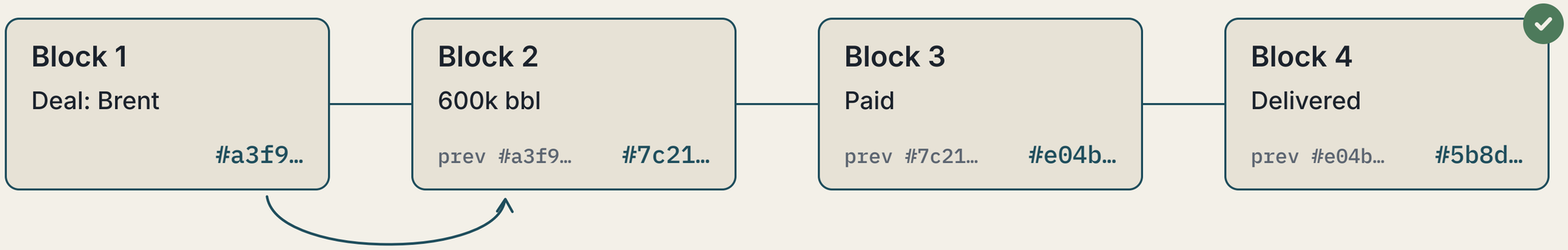
Seller

Buyer

Bank

Shipping

Inspector



add ✓ · secretly change ✗

Smart contract

IF	cargo arrives	✓
AND	passes inspection	✓
→ THEN	release payment	



Where it’s used



Upstream



Midstream



Downstream

Paying suppliers automatically

Pilot: 10 oil companies incl. Chevron, ExxonMobil, Shell

Platform: GumboNet

water-hauling trucks tracked → bills settled by smart contract

Every handover recorded once

Deals, documents, finance — digital

Origin & carbon footprint tracking

When does blockchain make sense?

Not for everything.

- ✓ Many companies share the same data
- ✓ They don’t fully trust each other
- ✓ Mistakes or fraud are expensive

Oil trading ✓✓✓



First market: North Sea crude (Brent)

VAKT

London · real-world example

Deal agreed → POST-TRADE: confirm · documents · finance · settle

2017–18	2019	2020
BP	Chevron	Saudi Aramco
Shell	Total	$5 M
Equinor	Reliance	

traders & banks

How a deal works

1 Enter trade

2 Match

3 Shipping & inspection

4 Finance

Built on Quorum

private Ethereum, first built by JPMorgan

Trade entered
#a3f9...

Matched
#7c21...

Inspection report
#e04b...

Financed
#5b8d...

only the parties in the deal can see it

The result





of North Sea oil deals on VAKT at launch (2019)

Still in use — 2024: Shell Trading Rotterdam adopts vSure

Challenges

Why it’s not everywhere yet

01 **Adoption**
works only if everyone joins



02 **Integration**
old trading & accounting systems

03 **Law & standards**
e-documents not accepted everywhere

04 **Privacy**
rivals must not see prices

05 **Garbage in, garbage out**
can’t verify data at entry



Benefits

One shared truth

no re-checking

Speed

days ⟶ minutes

Less fraud

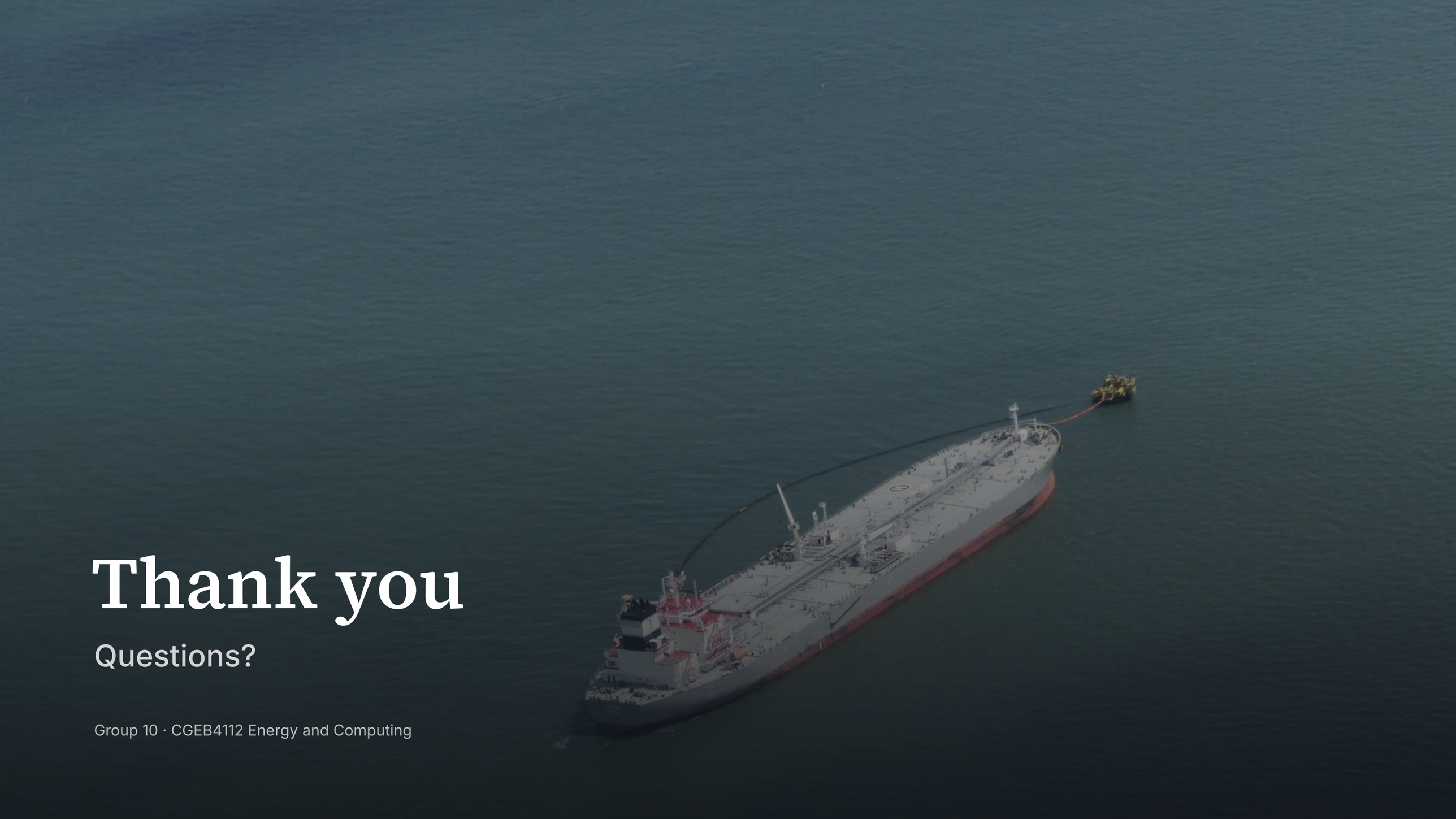
no double financing

Audit trail

who · what · when

Automation

smart contracts pay & approve



Thank you

Questions?

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